

Crossover study of diets enriched with virgin olive oil, walnuts or almonds. Effects on lipids and other cardiovascular risk markers.

BACKGROUND AND AIMS: Virgin olive oil (VOO) and nuts are basic components of the Mediterranean diet, a heart-healthy dietary pattern. Nuts have well known cholesterol lowering effects, while evidence is unclear for VOO. We designed a study in hypercholesterolemic patients to assess the effects on serum lipids and other intermediate markers of cardiovascular risk of replacing 40% of the fat in the background diet with VOO, walnuts or almonds. **METHODS AND RESULTS:** After a 4 week run-in period with a healthy diet, eligible candidates were randomized into three diet sequences in a crossover design, with a common background diet enriched with VOO, walnuts or almonds, lasting 4 weeks each. Outcomes were changes of serum lipids and oxidation and inflammation markers, measured by standard methods. Plasma fatty acids were determined by gas chromatography to assess compliance. In 18 participants completing the study (9 women, mean age 56 y, BMI 25.7 kg/m²), LDL-cholesterol was reduced from baseline by 7.3%, 10.8% and 13.4% after the VOO, walnut and almond diets, respectively ($P = 0.001$, Friedman test). Total cholesterol and LDL/HDL ratios decreased in parallel. LDL-cholesterol decreases were greater than predicted from dietary fatty acid and cholesterol exchanges among diets. No changes of other lipid fractions, oxidation analytes or inflammatory biomarkers were observed. Plasma fatty acid changes after each diet sequence supported good compliance. **CONCLUSION:** The results confirm the cholesterol lowering properties of nut-enriched diets. They also suggest that phenolic-rich VOO has a cholesterol lowering effect independently of its fatty acid content, which clearly deserves further study. Copyright © 2010 Elsevier B.V. All rights reserved.